

Practical – 04:

4(a)

Wait() ,waitpid()

```
akshitha_bushetty@LAPTOP-0FF1SH0T:~$ cd OSLab
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gedit practical4(a).c
-bash: syntax error near unexpected token `('
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gedit practical4.a.c

** (gedit:619): WARNING **: 04:21:06.683: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:619): WARNING **: 04:21:06.683: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gcc practical4.a.c -o practical4.a
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ./practical4.a
Parent Process: PID = 1545
Child 1: PID = 1546

Child 2: PID = 1547
Parent is using wait()...
Child 3: PID = 1548
Child 3 completed
wait(): Child with PID 1548 completed

Parent is using waitpid()...
Child 1 completed
waitpid(): Child 1 with PID 1546 completed
Child 2 completed
waitpid(): Child 2 with PID 1547 completed
waitpid(): Child 3 with PID 1548 completed

Parent: All child processes completed.
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ |
```

4(b):

Create a Zombie Process

```
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ./practical4.b
Parent Process: PID = 584
Parent is sleeping...
Child Process: PID = 585
Child is exiting...
Parent is exiting...
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ps -el
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
4 S 0 1 0 0 80 0 - 6081 - ? 00:00:00 system
5 S 0 2 1 0 80 0 - 795 - hvcc0 00:00:00 init-s
0 S 0 7 2 0 80 0 - 795 - hvcc0 00:00:00 init
4 S 0 47 1 0 79 -1 - 10556 - ? 00:00:00 system
4 S 989 79 1 0 80 0 - 5604 - ? 00:00:00 system
4 S 0 82 1 0 80 0 - 8871 - ? 00:00:00 system
1 S 0 160 1 0 80 0 - 39837 - ? 00:00:00 snapfu
1 S 0 161 1 0 80 0 - 39804 - ? 00:00:00 snapfu
1 S 0 168 1 0 80 0 - 39837 - ? 00:00:00 snapfu
1 S 0 176 1 0 80 0 - 39837 - ? 00:00:00 snapfu
1 S 0 178 1 0 80 0 - 39837 - ? 00:00:00 snapfu
1 S 0 189 1 1 80 0 - 113536 - ? 00:00:01 snapfu
1 S 0 193 1 0 80 0 - 39837 - ? 00:00:00 snapfu
4 S 0 220 1 0 80 0 - 722 - ? 00:00:00 chrony
4 S 0 221 1 0 80 0 - 1118 - ? 00:00:00 cron
4 S 995 222 1 0 80 0 - 2204 - ? 00:00:00 dbus-d
4 S 0 226 1 0 80 0 - 11024 - ? 00:00:00 networ
4 S 0 229 1 0 80 0 - 592181 - ? 00:00:00 snapd
4 S 0 231 1 0 80 0 - 4609 - ? 00:00:00 system
4 S 0 234 1 0 80 0 - 448075 - ? 00:00:00 wsl-pr
4 S 100 248 1 0 80 0 - 55137 - ? 00:00:00 rsyslo
4 S 0 263 1 0 80 0 - 1373 - tty1 00:00:00 agetty
4 S 0 280 1 0 80 0 - 30865 - ? 00:00:00 unatte
4 S 988 284 220 0 80 0 - 5106 - ? 00:00:00 chrony
1 S 988 300 284 0 80 0 - 3024 - ? 00:00:00 chrony
5 S 0 398 2 0 80 0 - 796 - ? 00:00:00 Sessio
5 S 0 399 398 0 80 0 - 800 - ? 00:00:00 RelayC
4 S 1000 404 399 0 80 0 - 1567 do_wai pts/0 00:00:00 bash
4 S 0 405 2 0 80 0 - 2162 - ? 00:00:00 login
4 S 1000 474 1 0 80 0 - 5698 - ? 00:00:00 system
5 S 1000 478 474 0 80 0 - 5730 - ? 00:00:00 (sd-pa
4 S 1000 505 405 0 80 0 - 1534 poll_s pts/1 00:00:00 bash
0 R 1000 597 404 0 80 0 - 1798 - pts/0 00:00:00 ps
```

4(c):

Remove the Zombie

```
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gedit practical4c.c
** (gedit:2439): WARNING **: 05:08:26.347: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2439): WARNING **: 05:08:26.347: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gcc practical4c.c -o practical4c
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ./practical4c
Parent Process: PID = 2809
Child Process: PID = 2810
Child is exiting...
Parent: Child process completed and collected.
Zombie process eliminated.
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ |
```